

Olympiad Test Paper — Science (Class 7)

Chapter 12: Reproduction in Plants

Subject	Science (NCERT Class 7)
Chapter	12 — Reproduction in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. All content stays within the NCERT Class 7 chapter on Reproduction in Plants — modes of reproduction, parts of a flower, pollination, fertilisation, fruit and seed formation, and seed dispersal. Many questions need two to four reasoning steps; watch for traps — especially the difference between *pollination* (transfer of pollen) and *fertilisation* (fusion of gametes). Do not write on this paper — mark answers on your answer sheet.

1. Which of the following is a feature of **asexual** reproduction?

- (A) only one parent is involved and no gametes fuse
(B) gametes fuse to form a zygote (C) two parents are always needed
(D) seeds are always produced

2. A gardener cuts a piece of a rose stem and plants it; it grows into a new rose plant. This is an example of:

- (A) vegetative propagation (B) sexual reproduction
(C) spore formation (D) fertilisation

3. New Bryophyllum plants grow from buds present on the:

- (A) tips of its roots (B) margins of its leaves
(C) inside of its flowers (D) surface of its seeds

4. The "eyes" of a potato are actually:

- (A) seeds formed after fertilisation (B) ovules waiting to be dispersed
(C) buds that can grow into new plants (D) tiny roots

5. Yeast reproduces by a process in which a small outgrowth forms on the parent and detaches. This is called:

- (A) budding (B) spore formation
(C) fragmentation (D) pollination

6. Spirogyra, a green alga, multiplies mainly by breaking into pieces, each piece growing into a new individual. This is:
- (A) fragmentation (B) budding
(C) vegetative propagation by leaf (D) cross-pollination
7. Bread mould (Rhizopus) and ferns both reproduce by:
- (A) budding (B) seed dispersal by wind
(C) stem cuttings (D) spore formation
8. Spores are especially useful to fungi because they:
- (A) can survive unfavourable conditions and spread easily by wind
(B) need two parents to form (C) are heavy and stay near the parent
(D) are produced only inside flowers
9. Which list contains **only** plants grown by vegetative propagation in everyday gardening?
- (A) potato, ginger, sugarcane, money plant (B) mango, coconut, balsam, maple
(C) yeast, Hydra, Spirogyra, Rhizopus (D) drumstick, madar, castor, lotus
10. A student says, "Plants grown from a cutting of one parent are exact copies of that parent." This statement is:
- (A) wrong, because cuttings always mix two parents' traits
(B) wrong, because cuttings produce seeds first
(C) correct, because asexual offspring are clones of the single parent
(D) correct only if the plant also flowers
11. The **male** reproductive part of a flower is the:
- (A) pistil (B) ovary
(C) stamen (D) stigma
12. A stamen is made up of:
- (A) stigma and style (B) petal and sepal
(C) ovary and ovule (D) anther and filament
13. Inside the flower, pollen grains are produced by the:
- (A) stigma (B) anther
(C) ovary (D) ovule
14. The pistil (carpel) of a flower consists of, from top to bottom:
- (A) stigma, style, ovary (B) anther, filament, ovary
(C) petal, sepal, ovule (D) ovary, style, anther
15. The female gametes of a flowering plant are found inside the:
- (A) anther (B) ovules
(C) filament (D) stigma

16. A flower that has **both** stamens and pistil is described as:

- (A) bisexual
- (B) unisexual
- (C) asexual
- (D) seedless

17. A papaya tree bears separate male flowers and female flowers. These flowers are best described as:

- (A) bisexual
- (B) spore-bearing
- (C) unisexual
- (D) self-pollinating

18. Pollination is correctly defined as the transfer of pollen from the:

- (A) ovary to the ovule
- (B) seed to the soil
- (C) stigma to the anther
- (D) anther to the stigma

19. The sticky, often feathery part of the flower on which pollen lands is the:

- (A) filament
- (B) stigma
- (C) anther
- (D) sepal

20. When pollen from a flower is carried to the stigma of **another flower on the same plant**, the pollination is:

- (A) self-pollination
- (B) cross-pollination
- (C) fertilisation
- (D) fragmentation

21. Cross-pollination occurs when pollen reaches the stigma of:

- (A) the same flower
- (B) a flower on a different plant of the same kind
- (C) another flower on the same plant
- (D) any plant of any kind

22. Which set of flower features would most suit pollination by **insects**?

- (A) dull colour, no scent, no nectar
- (B) bright colour, sweet scent, and nectar
- (C) very light, dry, powdery pollen and no petals
- (D) flowers hidden underwater

23. Wind-pollinated flowers typically have pollen that is:

- (A) sticky and heavy
- (B) wrapped in a hard fruit
- (C) brightly coloured and scented
- (D) light and produced in large amounts

24. A flower is small, green, scentless and produces huge amounts of light, dry pollen. Its likely pollinating agent is:

- (A) insects
- (B) water only
- (C) wind
- (D) birds attracted by colour

25. Fertilisation in a flowering plant is the:

- (A) transfer of pollen to the stigma
- (B) scattering of seeds by wind
- (C) ripening of the ovary into a fruit
- (D) fusion of the male gamete with the egg inside the ovule

- 26.** Which statement correctly separates pollination from fertilisation?
- (A) they are two names for the same event
 - (B) pollination needs water but fertilisation does not
 - (C) fertilisation happens first, then pollination
 - (D) pollination is the transfer of pollen; fertilisation is the fusion of gametes
- 27.** After a pollen grain lands on the stigma, the male gamete reaches the ovule by travelling through a:
- (A) pollen tube growing down the style
 - (B) hollow filament
 - (C) hole in the petal
 - (D) root hair
- 28.** Pollination must happen **before** fertilisation can occur because:
- (A) the ovary ripens into the fruit first
 - (B) pollen has to reach the stigma before the male gamete can fuse with the egg
 - (C) seeds are dispersed before the gametes meet
 - (D) the petals must fall off first
- 29.** The immediate product formed when the male gamete fuses with the female gamete is the:
- (A) fruit
 - (B) seed coat
 - (C) stigma
 - (D) zygote
- 30.** After fertilisation, the **ovary** of the flower develops into the:
- (A) seed
 - (B) pollen
 - (C) fruit
 - (D) new flower
- 31.** After fertilisation, each **ovule** develops into a:
- (A) fruit
 - (B) seed
 - (C) stamen
 - (D) petal
- 32.** A fruit is found to contain many seeds inside it. This tells us the flower's ovary had:
- (A) only one ovule
 - (B) many anthers
 - (C) no ovules
 - (D) many ovules
- 33.** After fertilisation, what usually happens to the petals, stamens and stigma of the flower?
- (A) they grow into the fruit
 - (B) they turn into seeds
 - (C) they become the ovary
 - (D) they wither and fall off
- 34.** Which sequence of events is in the correct order?
- (A) pollination → fertilisation → fruit and seeds form
 - (B) fertilisation → pollination → fruit forms
 - (C) seed dispersal → pollination → fertilisation
 - (D) fruit forms → pollination → fertilisation

35. A mango fruit develops from the flower's ovary, and inside it the single big stone (seed) develops from the:

- (A) anther
- (B) filament
- (C) stigma
- (D) ovule

36. Seed dispersal mainly helps plants by:

- (A) preventing overcrowding and spreading plants to new places
- (B) making seeds heavier
- (C) speeding up pollination
- (D) stopping fertilisation

37. If all the seeds of a plant simply fell and grew right below the parent, the seedlings would most suffer from:

- (A) too few of their own kind nearby
- (B) being unable to photosynthesise
- (C) competition for light, water, minerals and space
- (D) too much room to spread

38. Coconut seeds are well suited to dispersal by **water** because they are:

- (A) tiny and dry
- (B) covered with hooks
- (C) light, with a spongy, buoyant husk that floats
- (D) explosive when ripe

39. Drumstick and maple seeds have wing-like or hairy structures. These are adaptations for dispersal by:

- (A) water
- (B) wind
- (C) animals that eat them
- (D) explosion

40. Seeds of plants such as Xanthium that bear tiny hooks or spines are usually dispersed by:

- (A) bursting open suddenly
- (B) floating on water
- (C) being blown far by wind
- (D) sticking to the fur of animals

41. The pods of castor and balsam burst open when ripe and fling their seeds out. This is dispersal by:

- (A) wind
- (B) water
- (C) insects
- (D) explosion (sudden bursting)

42. The light, hairy seeds of the madar (Calotropis) plant are matched to which agent of dispersal?

- (A) wind
- (B) water
- (C) animals' digestive systems
- (D) explosion

43. A juicy berry is eaten by a bird; the hard seeds pass out unharmed far away. This describes seed dispersal by:

- (A) wind
- (B) animals
- (C) water
- (D) explosion

44. Which seed feature is **least** likely to help with wind dispersal?

- (A) being very light
- (B) having a hairy parachute
- (C) being heavy with a thick stony coat
- (D) having flat wings

45. A plant lives in still pond water and produces light, spongy floating fruits. The most suitable dispersal agent is:

- (A) wind through a forest canopy
- (B) animal fur
- (C) water
- (D) explosion

46. Reproduction by seeds (sexual reproduction) differs from vegetative propagation mainly because seeds:

- (A) result from two parents' gametes and so offspring can vary
- (B) are formed without any flower
- (C) always produce exact clones of one parent
- (D) cannot be dispersed

47. A farmer wants every new sugarcane plant to be identical to one prized parent plant. The best method is to:

- (A) collect and sow its seeds
- (B) wait for wind pollination
- (C) cross-pollinate it with another variety
- (D) plant stem cuttings from that parent

48. Hydra, a tiny water animal, develops a small bulge that grows and separates into a new individual. This same mode of reproduction is also seen in:

- (A) potato
- (B) ferns
- (C) yeast
- (D) coconut

49. Which of these correctly pairs an organism with its mode of asexual reproduction?

- (A) Rhizopus (bread mould) - spore formation
- (B) Spirogyra - spore formation
- (C) yeast - fragmentation
- (D) potato - budding

50. A money-plant cutting placed in a glass of water grows roots and becomes a new plant. This shows that vegetative propagation can occur from a:

- (A) seed
- (B) spore
- (C) stem
- (D) pollen grain

51. Which statement about pollen is correct?

- (A) pollen is produced inside the ovary
- (B) pollen becomes the seed after dispersal
- (C) pollen carries the male gamete and is made in the anther
- (D) pollen lands on the filament during pollination

52. In a bisexual flower, self-pollination is possible because:

- (A) it has flowers on two different plants
- (B) it has no pistil
- (C) it produces spores instead of pollen

(D) both the anther (pollen) and the stigma are present in the same flower

53. A drumstick seed has papery wings; a coconut has a spongy husk. The reason their structures differ is that they are adapted to:

- (A) the same dispersal agent
- (B) different dispersal agents - wind for one, water for the other
- (C) animal dispersal in both cases
- (D) explosive dispersal in both cases

54. Why can a banana plant produce fruit even though its cultivated variety makes no usable seeds?

- (A) it reproduces only by spores
- (B) its seeds are dispersed by wind before we see them
- (C) it is grown by vegetative propagation from its stem, not from seeds
- (D) it does not need to reproduce at all

55. A flower is pollinated successfully, but no fertilisation follows and no fruit forms. The most reasonable explanation is that:

- (A) the ovary turned straight into seeds
- (B) the male gamete never fused with the egg in the ovule
- (C) the petals failed to wither
- (D) the stigma rejected the stamen

56. Place these in the correct life-cycle order: (i) seed germinates into a new plant, (ii) flower is pollinated, (iii) fruit with seeds forms, (iv) fertilisation occurs.

- (A) (i) → (ii) → (iii) → (iv)
- (B) (ii) → (iv) → (iii) → (i)
- (C) (iv) → (ii) → (i) → (iii)
- (D) (iii) → (i) → (ii) → (iv)

57. Which observation would best show that a plant reproduced **sexually** rather than by vegetative propagation?

- (A) the new plant grew from a buried stem tuber
- (B) the new plant grew from a stem cutting
- (C) the new plant grew from a leaf-margin bud
- (D) the new plant grew from a seed formed inside a fruit

58. A bee crawls into a bright, scented flower to drink nectar and leaves dusted with pollen, which it carries to the next flower. The bee is acting as:

- (A) an agent of fertilisation
- (B) an agent of pollination
- (C) an agent of seed dispersal
- (D) a parent in vegetative propagation

59. Two strawberry varieties are deliberately cross-pollinated to combine their best traits in the offspring. This deliberate crossing is called:

- (A) fragmentation
- (B) hybridisation
- (C) budding
- (D) self-pollination

60. A gardener notices new plants sprouting directly from the underground stem of a ginger plant, with no flowers, seeds or pollen involved anywhere. The gardener should

conclude that the ginger reproduced by:

- (A) sexual reproduction through pollination
- (B) seed dispersal by animals
- (C) vegetative propagation (asexual reproduction)
- (D) spore formation

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 12:

Reproduction in Plants

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	B	31	B	41	D	51	C
2	A	12	D	22	B	32	D	42	A	52	D
3	B	13	B	23	D	33	D	43	B	53	B
4	C	14	A	24	C	34	A	44	C	54	C
5	A	15	B	25	D	35	D	45	C	55	B
6	A	16	A	26	D	36	A	46	A	56	B
7	D	17	C	27	A	37	C	47	D	57	D
8	A	18	D	28	B	38	C	48	C	58	B
9	A	19	B	29	D	39	B	49	A	59	B
10	C	20	A	30	C	40	D	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (A)** Asexual reproduction needs only one parent and involves no fusion of gametes (no fertilisation); the offspring are clones. Two parents and gamete fusion belong to sexual reproduction, and seeds are not always produced (e.g. a stem cutting).
- (A)** Growing a whole new plant from a piece of stem of one parent — with no flower, pollen or seed — is vegetative propagation, a kind of asexual reproduction.
- (B)** Bryophyllum is the classic example: tiny buds form along the notches at the margins of its leaves, and each can grow into a new plant.
- (C)** The 'eyes' of a potato are buds on the swollen underground stem (tuber); planting a piece with an eye grows a new plant. They are not seeds or roots.
- (A)** In yeast, a small outgrowth (bud) forms on the parent, grows, and detaches to become a new yeast cell — this is budding (also seen in Hydra).
- (A)** Spirogyra breaks (fragments) into pieces, and each piece grows into a new filament. This mode is fragmentation.

- 7. (D)** Both bread mould (*Rhizopus*) and ferns reproduce by spore formation — tiny spores that spread and germinate into new individuals.
- 8. (A)** Spores have a tough protective coat, so they can survive unfavourable conditions, and being light they are carried far and wide by wind — both are why spores are so successful for fungi and ferns.
- 9. (A)** Potato (tuber buds), ginger (underground stem), sugarcane (stem cuttings) and money plant (stem cuttings) are all everyday examples of vegetative propagation. The other lists mix in seed-bearing plants or budding/fragmenting organisms.
- 10. (C)** In asexual reproduction there is a single parent and no mixing of gametes, so the offspring are genetically identical — clones. Hence the statement is correct, and cuttings do not first make seeds.
- 11. (C)** The stamen is the male reproductive part of the flower (it makes pollen). The pistil/carpel, with its ovary and stigma, is the female part.
- 12. (D)** A stamen has two parts: the anther (which makes pollen) sitting on top of a stalk called the filament.
- 13. (B)** Pollen grains, which carry the male gametes, are produced in the anther of the stamen.
- 14. (A)** The pistil (carpel) is the female part: from top to bottom it is the stigma, then the style, then the ovary (which holds the ovules).
- 15. (B)** The female gametes (eggs) are inside the ovules, which lie within the ovary of the pistil.
- 16. (A)** A flower that carries both stamens (male) and a pistil (female) is bisexual. If it had only one of them it would be unisexual.
- 17. (C)** Separate male and female flowers means each flower has only one kind of reproductive part, so these flowers are unisexual (as in papaya).
- 18. (D)** Pollination is, by definition, the transfer of pollen from the anther (where it is made) to the stigma (where it lands).
- 19. (B)** The stigma, the sticky and often feathery top of the pistil, is where pollen lands. The anther produces pollen and the filament holds the anther up.
- 20. (A)** Pollen reaching the stigma of the SAME plant — even a different flower on that plant — is still self-pollination. Cross-pollination needs a different plant of the same kind.

- 21. (B)** Cross-pollination is when pollen reaches the stigma of a flower on a DIFFERENT plant of the same kind. Same flower or same plant would be self-pollination.
- 22. (B)** Insects are drawn by sight, smell and food, so insect-pollinated flowers tend to be brightly coloured and sweetly scented, with nectar as a reward.
- 23. (D)** Wind cannot aim, so wind-pollinated flowers make pollen that is light and in large amounts, raising the chance that some lands on a stigma. Sticky/heavy or showy pollen suits insects.
- 24. (C)** Small, green, scentless flowers offer no attraction to insects; producing huge amounts of light, dry pollen is the give-away that the agent is wind.
- 25. (D)** Fertilisation is the fusion of the male gamete (from the pollen) with the female gamete (egg) inside the ovule, forming a zygote. Pollen transfer and fruit ripening are separate steps.
- 26. (D)** This is the classic trap: pollination is only the TRANSFER of pollen to the stigma; fertilisation is the later FUSION of the male and female gametes. They are different events, pollination first.
- 27. (A)** After landing on the stigma, the pollen grain grows a pollen tube down through the style to carry the male gamete to the ovule.
- 28. (B)** Fertilisation needs the male gamete (carried by pollen) to reach the egg; that gamete cannot get there until pollen has first been transferred to the stigma. So pollination must come before fertilisation.
- 29. (D)** The fusion of the male and female gametes forms a single cell called the zygote, which later develops into the embryo of the seed.
- 30. (C)** After fertilisation, the ovary ripens and develops into the fruit (which encloses the seeds).
- 31. (B)** After fertilisation, each ovule inside the ovary develops into a seed. (The ovary itself becomes the fruit.)
- 32. (D)** Each seed comes from one ovule, so a fruit with many seeds must have formed from an ovary that contained many ovules.
- 33. (D)** Once fertilisation is done, the parts that are no longer needed — the petals, stamens and stigma — usually wither and fall off, while the ovary grows into the fruit.
- 34. (A)** The correct order is: pollination (pollen reaches stigma) → fertilisation (gametes fuse) → fruit and seeds form. The other orders reverse or misplace these steps.

- 35. (D)** The fruit forms from the ovary, while the seed (the mango stone) forms from the ovule inside that ovary.
- 36. (A)** Dispersal carries seeds away from the parent, which prevents overcrowding and lets the plant colonise new places where there is room, light, water and minerals.
- 37. (C)** If seeds all grew right under the parent, the seedlings would crowd together and fight one another (and the parent) for light, water, minerals and space — exactly what dispersal avoids.
- 38. (C)** A coconut floats because of its light, fibrous, spongy and buoyant husk, so it is well suited to dispersal by water (it can even cross the sea).
- 39. (B)** Wings (maple) and hairs (drumstick) let seeds catch the air and be carried far, so these are adaptations for dispersal by wind.
- 40. (D)** Tiny hooks or spines catch on the fur of passing animals, so the seeds are carried away stuck to the animal — dispersal by animals.
- 41. (D)** Castor and balsam fruits dry, build up tension, and suddenly burst open, flinging seeds away from the parent — dispersal by explosion (sudden bursting).
- 42. (A)** Madar (Calotropis) seeds are very light and bear silky hairs that act like parachutes, so they are carried by wind.
- 43. (B)** When an animal eats a juicy fruit, the tough seeds pass out unharmed in its droppings far from the parent — dispersal by animals.
- 44. (C)** Wind dispersal relies on seeds being light, often with hairs or wings to catch the air. A seed that is heavy with a thick stony coat is the least suited to being blown about.
- 45. (C)** A still-water plant with light, spongy floating fruits is adapted for dispersal by water, which carries the floating fruit to new spots.
- 46. (A)** Seeds form from the fusion of gametes from (usually) two parents, so the offspring can vary and differ from the parents. Vegetative propagation, in contrast, makes identical clones from one parent.
- 47. (D)** To get plants identical to one prized parent, the farmer should use vegetative propagation — stem cuttings — because asexual offspring are clones. Seeds or cross-pollination would introduce variation.
- 48. (C)** Hydra reproduces by budding (a bulge grows and separates). Among the choices, yeast also reproduces by budding.

- 49. (A)** Bread mould (*Rhizopus*) reproduces by spore formation — the correct pairing. *Spirogyra* fragments, yeast buds, and the potato propagates vegetatively, so the other pairings are wrong.
- 50. (C)** A money-plant cutting is a piece of STEM that grows roots and becomes a new plant, showing vegetative propagation from a stem.
- 51. (C)** Pollen carries the male gamete and is produced in the anther. It is not made in the ovary, does not become the seed, and lands on the stigma (not the filament).
- 52. (D)** A bisexual flower has both the pollen-bearing anther and the stigma in the same flower, so pollen can reach its own stigma — self-pollination is possible.
- 53. (B)** Different seed structures match different agents: a drumstick's papery wings suit wind, while a coconut's spongy husk suits water. So they are adapted to different dispersal agents.
- 54. (C)** Cultivated bananas are seedless, yet they keep being grown because new banana plants are raised by vegetative propagation from the parent (its stem/sucker), not from seeds.
- 55. (B)** Pollination (pollen on the stigma) does not guarantee fertilisation. If the male gamete never fuses with the egg in the ovule, there is no zygote, so no seed and no fruit form.
- 56. (B)** The life-cycle order is: the flower is pollinated (ii), fertilisation occurs (iv), a fruit with seeds forms (iii), and a seed then germinates into a new plant (i). So (ii) → (iv) → (iii) → (i).
- 57. (D)** Sexual reproduction makes seeds (from fertilisation) inside fruits, so a plant grown from a seed formed in a fruit shows sexual reproduction. Tubers, leaf-buds and cuttings are all vegetative (asexual).
- 58. (B)** The bee carries pollen from one flower to another's stigma, so it is acting as an agent of pollination. (Fertilisation is the later fusion of gametes, which the bee does not perform.)
- 59. (B)** Deliberately cross-pollinating two different varieties to combine their good traits in the offspring is called hybridisation.
- 60. (C)** New plants growing straight from the underground stem, with no flower, seed or pollen, is vegetative propagation — a form of asexual reproduction.